Resume Text Analytics for Intelligent College Placement Support

A Project Based Learning Report Submitted in partial fulfilment of the requirements for the award of the degree

of

Bachelor of Technology

TEXT ANALYTICS

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Submitted by

2310080022: JAGGAVARAPU AKHIL SAI REDDY

2310080007: MOHAMMED SHOAIB HANNAN

Under the guidance of

MR. MANI KUMAR CHAKILAM

Department of Artificial Intelligence and Data science

Koneru Lakshmaiah Education Foundation, Aziz Nagar

Aziz Nagar – 500075

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Introduction

In today's highly competitive job market, the college placement process plays a vital role in shaping students' professional careers and future opportunities. With a significant increase in the number of students participating in campus recruitment drives, managing and evaluating large volumes of resumes has become a complex and time-intensive task for placement cells and recruiters. Traditional resume screening methods rely heavily on manual effort, which not only consumes considerable time but also introduces inconsistencies and the possibility of human bias. As a result, deserving candidates may be overlooked, and the overall efficiency of the recruitment process is affected.

To overcome these challenges, Resume Text Analytics has emerged as an intelligent and automated solution. This approach leverages advanced techniques from Natural Language Processing (NLP) and Machine Learning (ML) to systematically analyze, interpret, and process resume data. Since resumes are inherently unstructured and vary in format and content, extracting meaningful information such as skills, educational background, projects, certifications, and work experience requires sophisticated text processing and understanding. Resume text analytics effectively transforms this unstructured data into structured and standardized information, enabling accurate evaluation and comparison across multiple candidates. The proposed system aims to develop a smart and efficient platform that assists both students and placement coordinators in streamlining the placement process. It automates resume parsing, extracts relevant features, and applies intelligent algorithms to match candidate profiles with job requirements. By doing so, the system significantly reduces the time required for screening while ensuring consistency and fairness in evaluation. Additionally, it can rank candidates based on their suitability for specific roles, thereby helping recruiters make quicker and more informed decisions.

Another important aspect of the system is its ability to provide personalized insights and feedback to students. By identifying skill gaps and highlighting areas for improvement, the system guides students in enhancing their resumes and aligning their profiles with industry expectations. This not only improves individual employability but also increases the overall placement success rate of the institution.

Moreover, the system supports data-driven decision-making by maintaining a centralized database of student profiles and analytics. Placement cells can use this data to track trends, identify in-demand skills, and design targeted training programs to better prepare students for recruitment. The integration of modern web technologies further ensures that the system is user-friendly, scalable, and accessible.

In addition, the adoption of such an automated system promotes transparency and minimizes subjective judgment in the recruitment process. By applying uniform evaluation criteria to all candidates, it ensures equal opportunities and a fair selection process. Overall, Resume Text Analytics represents a significant step towards modernizing college placement systems by making them faster, more accurate, efficient, and reliable, ultimately benefiting both students and recruiters.

LITERATURE REVIEW

The field of Resume Text Analytics has experienced rapid growth with the advancement of Natural Language Processing (NLP) and Machine Learning (ML), significantly transforming the way recruitment and placement processes are conducted. Early research in this domain primarily focused on automating resume parsing, where systems aimed to extract essential information such as skills, educational qualifications, work experience, and contact details from unstructured resume documents [1]. These initial systems largely depended on rule-based approaches, regular expressions, and keyword matching techniques to identify and segment different sections of resumes [4]. Although effective to some extent, these methods lacked flexibility and struggled with variations in resume formats and writing styles.

As research progressed, more sophisticated NLP-based techniques were introduced to improve the accuracy and adaptability of resume parsing systems. Techniques such as tokenization, part-of-speech tagging, and syntactic parsing enabled systems to better understand the grammatical structure of resume content [7]. Studies demonstrated that NLP-driven parsers could convert unstructured resumes into structured representations such as JSON or database entries, thereby facilitating efficient storage, retrieval, and analysis [1]. This marked a significant step toward building intelligent recruitment systems capable of handling large-scale data.

Subsequent research shifted focus from simple information extraction to intelligent resume screening and candidate evaluation. Researchers explored various similarity-based approaches, including cosine similarity, Jaccard similarity, and TF-IDF (Term Frequency–Inverse Document Frequency), to measure the relevance between candidate resumes and job descriptions [10,11,12]. These techniques enabled automated matching of candidates to job roles by calculating similarity scores based on textual features [3]. Such systems improved recruitment efficiency by reducing manual effort, accelerating the shortlisting process, and ensuring more consistent evaluation criteria. In the context of college placement systems, these methods proved particularly valuable due to the high volume of resumes that need to be processed within limited timeframes.

With the evolution of artificial intelligence, resume analytics has advanced beyond traditional machine learning techniques to incorporate deep learning and transformer-based architectures. Models such as Bidirectional Encoder Representations from Transformers (BERT) and its variants have significantly enhanced the contextual understanding of textual data [6]. Unlike earlier models, these approaches capture semantic relationships and contextual meanings within resumes, leading to more accurate extraction of information and improved matching with job requirements. Additionally, the use of word embeddings and contextual representations has enabled systems to recognize synonyms and related skills, thereby overcoming limitations of keyword-based matching [5].

Another important advancement in this domain is the application of Named Entity Recognition (NER). NER techniques are used to identify and classify key entities within resumes, such as technical skills, programming languages, organizations, job roles, certifications, and educational institutions [9]. By accurately extracting and categorizing these entities, NER-based systems improve the precision and reliability of structured resume data. This structured information serves as a strong foundation for downstream tasks such as candidate ranking, recommendation, and analytics.

Recent studies have also explored the integration of Optical Character Recognition (OCR) with NLP techniques to process resumes in various formats, including scanned documents and images [4]. This enhances system flexibility and ensures that resumes in non-editable formats can also be analyzed effectively. Furthermore, hybrid systems combining rule-based methods, machine learning models, and deep learning techniques have demonstrated improved performance in handling diverse resume structures.

In addition to extraction and matching, modern research emphasizes candidate ranking and recommendation systems. Advanced algorithms are used to rank candidates based on multiple factors such as skill relevance, experience level, and job compatibility [3]. Recommendation systems further enhance functionality by suggesting suitable job roles, career paths, or skill development opportunities for candidates. These systems often utilize collaborative filtering and content-based filtering techniques to generate personalized recommendations.

Another emerging area in resume analytics is the use of predictive analytics, where models are trained to predict candidate success or job performance based on historical data. Such approaches provide valuable insights to recruiters and help institutions design targeted training programs to improve student employability. Moreover, fairness-aware machine learning techniques are being explored to address issues related to bias and ensure equitable evaluation across all candidates [4].

Overall, the literature indicates that Resume Text Analytics has evolved from basic rule-based parsing systems to highly advanced, AI-driven solutions capable of intelligent decision-making. The integration of NLP, machine learning, deep learning, and data analytics has made these systems more scalable, accurate, and efficient. These advancements are particularly relevant in the context of college placement support systems, where they not only automate the recruitment process but also provide actionable insights to students and institutions. Consequently, Resume Text Analytics plays a crucial role in bridging the gap between student capabilities and industry requirements, making the placement process more effective, transparent, and data-driven.

Methodology

The proposed system, PlacementBot – Resume Text Analytics for Intelligent Placement Support, follows a structured pipeline integrating frontend interfaces, backend processing, and intelligent analytics to automate and enhance the placement process.

User Interaction & Data Input

The system begins with a user-friendly web interface built using React, where students can interact with multiple modules such as Dashboard, AI Chatbot, Resume Upload, Analytics, and Profile. Students upload resumes through a drag-and-drop interface or file browser, supporting formats like PDF and DOCX. User details are managed through a profile module, ensuring personalized analysis.

Resume Upload & Storage

Once a resume is uploaded via the Resume Analyzer module, it is securely transferred to the backend server. The system stores the resume along with metadata (upload date, ATS score, etc.) in a MongoDB database, enabling efficient storage and retrieval of multiple resumes per user.

Resume Parsing & Text Extraction

The uploaded resumes are processed using parsing libraries to extract textual content. For standard documents (PDF/DOCX), text extraction tools are used, while OCR (Optical Character Recognition) is applied for scanned or image-based resumes. This step converts raw resume files into machine-readable text for further processing.

Text Preprocessing

The extracted text is cleaned and normalized using NLP preprocessing techniques such as lowercasing text, removing stop words and special characters, tokenization, and basic normalization. Libraries such as NLTK or spaCy are used to ensure high-quality, structured input for analysis.

Skill & Information Extraction

The system extracts key information such as technical skills, domains, projects, and experience using techniques like keyword extraction and Named Entity Recognition (NER). The extracted data is displayed in the dashboard (e.g., “Your Top Skills”) and stored for analytics.

ATS Scoring & Resume Evaluation

The system evaluates resumes using an ATS (Applicant Tracking System) scoring mechanism that considers resume structure, keyword relevance, and skill-job alignment. A score is generated and displayed along with a detailed analysis summary explaining strengths and weaknesses.

Job Role Prediction & Matching

Based on extracted features, the system predicts suitable job roles. Similarity techniques such as cosine similarity are used to compare resumes with job requirements, ensuring accurate matching.

Feedback & Improvement Suggestions

The system provides personalized feedback by identifying missing or weak skills and suggesting improvements such as adding measurable achievements or certifications. These insights help students enhance their resumes.

Analytics & Visualization

An advanced analytics module provides visual insights such as skill distribution, resume performance scores, learning roadmaps, and growth opportunities. This helps students understand their strengths and areas for improvement.

AI Chatbot Assistance

The system includes an AI-powered chatbot that provides placement guidance, interview preparation tips, and answers to career-related queries, improving user engagement and support.

Ranking & Decision Support

Candidates are ranked based on ATS scores, skills, and matching results. This helps placement officers quickly identify top candidates and ensures fair evaluation.

Frontend–Backend Integration

The system uses REST APIs to connect the React frontend with backend services. Resumes uploaded from the frontend are processed in the backend, analyzed using NLP and ML techniques, stored in MongoDB, and results are sent back to the user interface.

System Output

The final output is displayed through an interactive dashboard and includes ATS score, extracted skills, resume insights, job recommendations, personalized feedback, and analytics. This supports both students and placement officers in making informed decisions.

System Architecture

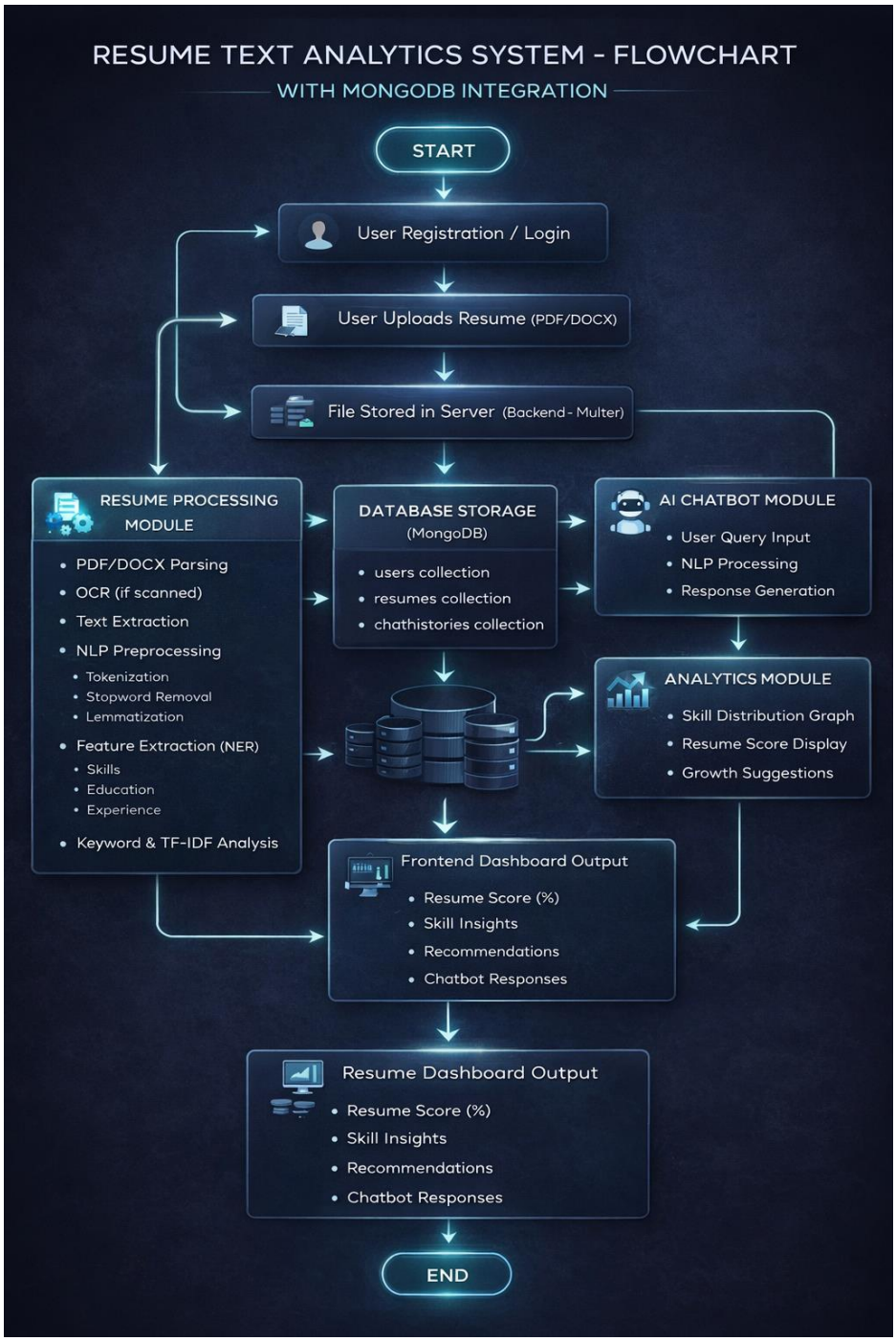

Figure 1: Detailed System Architecture of Resume Text Analytics System

This figure represents the complete workflow of the Resume Text Analytics system, starting from user login and resume upload to processing using NLP techniques. It illustrates how data is stored in MongoDB and utilized by modules like analytics and AI chatbot. The system finally generates outputs such as resume scores, skill insights, and recommendations through the dashboard.

Outcomes of the Project

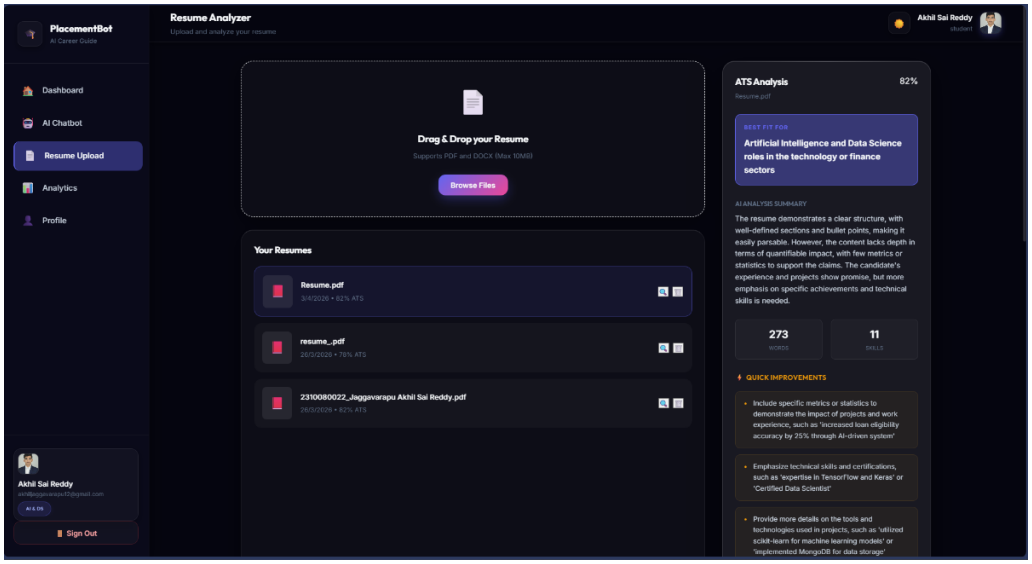

Figure 2.1: User Interface for Resume Upload and ATS Analysis

This figure shows the user interface of the Resume Analyzer system, where users can upload resumes and view analysis results. It includes features such as ATS score evaluation, resume listing, and improvement suggestions. The dashboard provides an intuitive way to manage and analyze resumes efficiently.

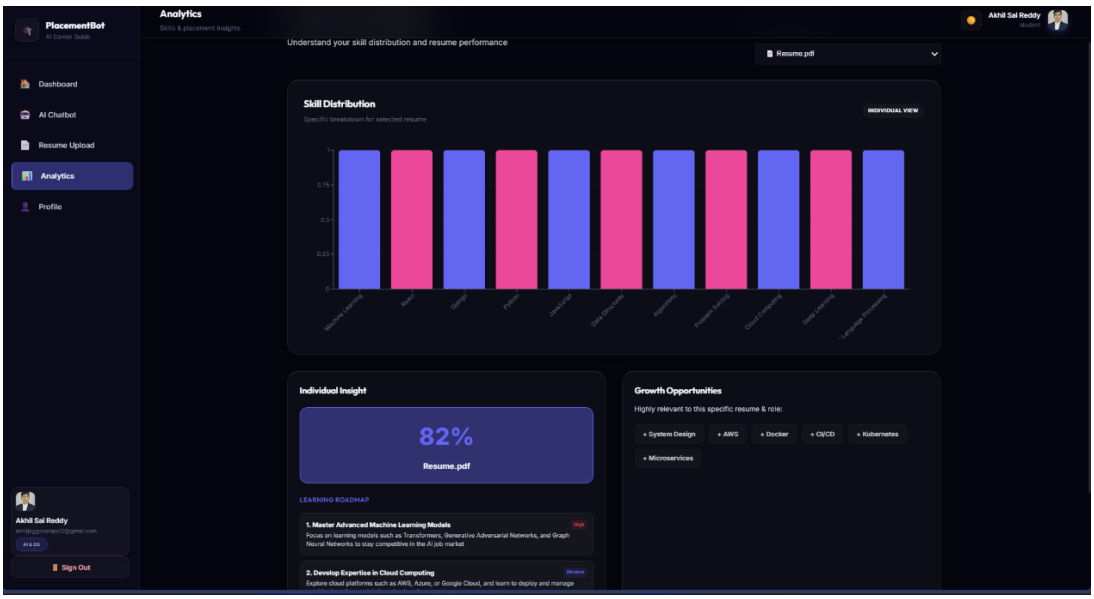

Figure 2.2: Analytics Dashboard Showing Skill Distribution and Resume Insights

This figure illustrates the analytics dashboard of the Resume Text Analytics system, displaying skill distribution and performance insights. It highlights the resume score, individual analysis, and recommended growth opportunities. The dashboard helps users understand their strengths and identify areas for skill improvement.

Project Impact & Future Scope

1. Current Impact

The developed Resume Text Analytics system has a significant impact on improving the efficiency and effectiveness of the college placement process. The key contributions include:

24/7 Instant Career Guidance:

The integrated AI chatbot provides continuous support to students by offering placement guidance, interview tips, and career advice at any time.

Data-Driven Resume Optimization:

The system analyzes resumes and generates ATS scores along with improvement suggestions, helping students enhance their resumes to meet industry standards and increase their chances of selection.

Personalized Learning Pathways:

By identifying skill gaps and analyzing industry requirements, the system recommends relevant skills and technologies, enabling students to bridge the gap between academic knowledge and industry expectations.

Improved Placement Efficiency:

Automation of resume screening reduces manual effort for placement cells and ensures faster, more consistent candidate evaluation.

2. Future Scope

The system has strong potential for further enhancement with advanced features and integrations:

AI-Based Mock Interviews:

Integration of real-time voice and video-based interview simulations using AI to help students practice and improve their communication and problem-solving skills.

Job Aggregator Integration:

Automatic matching of analyzed resumes with live job postings from multiple job portals, enabling students to discover relevant opportunities instantly.

Community Collaboration Platform:

Development of a peer-to-peer portal where students can share interview experiences, discuss preparation strategies, and learn collaboratively.

Advanced AI Models:

Incorporation of deep learning and transformer-based models for more accurate resume understanding and job matching.

Industry Integration:

Direct collaboration with companies to align the system with real-world hiring requirements and improve placement outcomes.

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